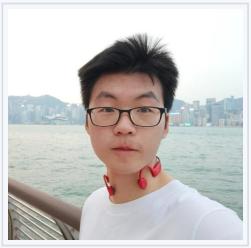
RESEARCH CV LLMs & AGENTS

Zhuohang Jiang

LLMs · AI Agents · Multimodal RAG · Reasoning

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Research Profile

LLM reasoning, multimodal RAG, self-evolving agents. **8 published/accepted contributions; 7 at CCF-A venues/related tracks, 5 first/co-first authored**: KDD, CVPR, NeurIPS, AAAI.

Education

The Hong Kong Polytechnic University | Ph.D. in Computer Science Sep. 2024 – Present
Advisors: Prof. Qing Li and Prof. Wenqi Fan; GPA: **3.60/4.00**.

Sichuan University | B.Eng. in Computer Science and Technology Sep. 2020 – Jun. 2024
Major GPA: **3.79/4.00**; overall GPA: 3.78/4.00.

Industry Experience

Kuaishou Technology | Research Intern, Short-Video E-commerce Oct. 2025 – Present

- Developed **AB Agent**: structured strategy/code changes, parameters, and outcomes into retrievable units; integrated A/B comparisons, tuning recommendations, configuration checks, and review. Achieved **GMV +4.829%** with positive guardrails and continuous experience updates.
- Developed **AIR**: decomposed LLM-derived intent semantics into reusable atomic knowledge; approximated sequence-level reasoning through online retrieval/composition, reducing real-time LLM costs.

Selected Research

HiBench | Hierarchical Structure Reasoning in LLMs

First Author / Project Lead · KDD 2025, Benchmark & Dataset Track

Implementation: Led benchmark design and code development; controlled depth/width/node count for **tree/JSON/formula generators**, algorithmic labels, and prompts; added code/paper tasks (**30 tasks, 39,519 queries**). **Evaluation:** Compared representations, complexity, and CoT across **20 LLMs / 10 families**. Instruction tuning yielded relative gains of **88.84% (Llama-3.1-8B)** and **31.38% (Qwen2.5-7B)**; released data/toolkit.

SUPERGLASSES / SUPERLENS | Smart-Glasses VLM Benchmark & Agent

First Author · CVPR 2026 Findings

Data: Collected **2,422 real-device QA pairs / 14 domains / 8 query types** with tool-use traces. **Agent:** Routed direct answers vs. RAG; localized small objects via open-vocabulary detection and decomposed multi-hop queries. Searched image/text evidence, cleaned HTML, and reranked chunks; **two-layer Redis caching** reuses query-to-URL and URL-to-content results. Evaluated **26 VLM/RAG variants**; SUPERLENS improved Qwen2.5-VL-7B from **32.82 to 44.10 (+11.28 points)**.

A/B Agent | Self-Evolving Agent for Industrial Strategy Iteration

First Author · arXiv:2608.04625, 2026 (Preprint)

Tree-RAG: Indexed experience by domain/scenario/ranking stage/objective. Fused **sparse/dense retrieval**, tree-path proximity boosts, and **Qwen-Reranker**; generated evidence-backed configurations and screened feasibility/guardrail risks. **Self-evolution:** Stored strategies, parameters, metrics, and context in experiment-tree nodes; compared **parent–child/sibling trials**, confidence-weighted gains, and guardrail penalties to guide local parameter search or switch strategy branches.

AIR | Composable LLM Intent Reasoning with Offline-to-Online Retrieval

First Author · KDD 2026, Applied Data Science Track

LLM reasoning: Mapped behavior, user attributes, and object descriptions into hierarchical intent paths, cached as reusable atomic knowledge. **Retrieval/composition:** Built intent trees with action weights/time decay; retrieved target-relevant chains and fused evidence via **multi-head attention**, approximating sequence-level semantics without online LLM calls. Achieved approximately **400× throughput**; e-commerce deployment yielded **GMV +3.446%**.

QA-Dragon | Query-Aware Dynamic RAG for Knowledge-Intensive VQA

First Author · KDD Cup 2025 Workshop on Multimodal Retrieval Augmented Generation

Routed drafts to **direct output / search verification / RAG**; selected image/text tools, cropped target objects, and reranked evidence before answer verification. Single-source/multi-source/multi-turn gains: **5.06% / 6.35% / 5.03%**.

Academic Activities, Honors & Skills

Activities: KDD 2025 Reasoning Day invited talk; KDD Cup 2025 CRAG-MM, 12th globally (team lead); NeurIPS 2024 Spotlight (co-first author).
Skills: PyTorch, Hugging Face, RAG, agent systems, Docker, Git; Mandarin (native), English (IELTS 6.5).